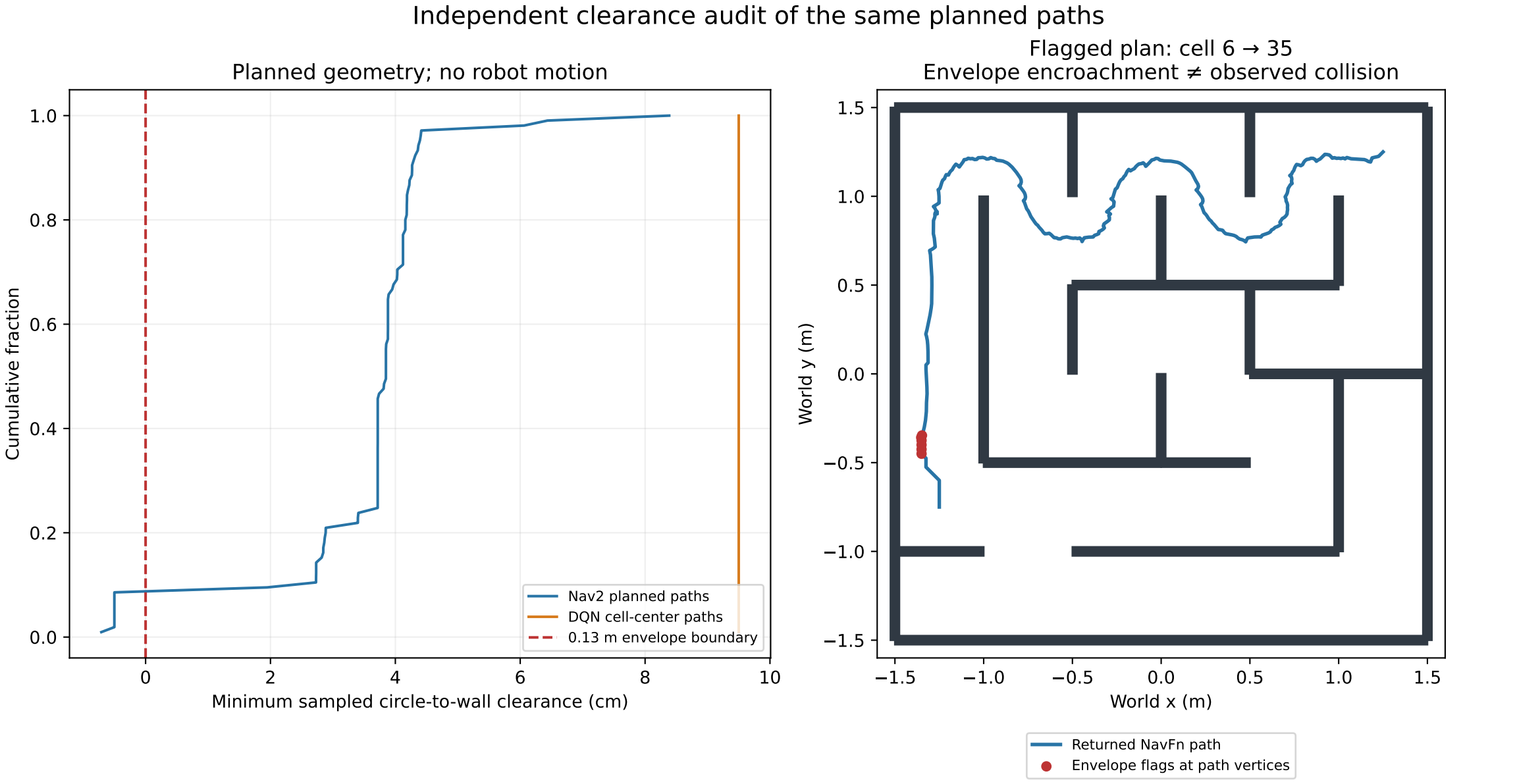
Independent clearance audit of the same planned paths
Planned geometry; no robot motion
Cumulative fraction
Minimum sampled circle-to-wall clearance (cm)
Nav2 planned paths
DQN cell-center paths
0.13 m envelope boundary
Flagged plan: cell 6 → 35
Envelope encroachment ≠ observed collision
World y (m)
World x (m)
Returned NavFn path
Envelope flags at path vertices